

Scenario 1: Imagine a market researcher surveys 45 people on their breakfast-eating habits. She wants to know whether people in the local region eat breakfast every morning before leaving their home.

1. In this scenario, what is the sample?

45 people surveyed

2. In this scenario, what is the population?

People in the local region

Scenario 2: Brenda wants to collect data on the favorite music genre by young people in Titusville, FL. She decides to go outside a local music store and ask every person who comes into the store during the morning and afternoon about their music preferences.

3. What is the population?

Young people in Titusville, FL

4. What is the sample?

People who go to the local music store during morning & afternoon

5. Explain why a sample chosen in this way may not be representative of the population.

People who go to the music store may not be young & would include all ages.

Scenario 3: Morgan is going to sell cookies outside of her house to raise money for her troop. She wants to predict her sales and estimate how many days she must sell the cookies. She is considering three different methods for surveying a sample. Describe the benefits and problems with each of the possible survey methods.

Answers may vary.

6. Ask everyone on her class if they like cookies and how many will they eat in a day.

A benefit is that her classmates have a variety of wants.

A problem is that her classmates are not a representative sample of people she is selling cookies to outside of her house.

7. Ask all people younger than 15 around her house if they like cookies and how many will they eat in a day.

A benefit is that there will be a variety of wants.

A problem is again this is not representative of the population of all ages buying cookies.

8. Ask all neighbors around her house if they like cookies and how many will they drink in a day.

A benefit is that her neighbors would be more representative of the population of people buying cookies.

A problem is that it doesn't incorporate people passing by who are not neighbors.

9. Explain which of the methods from questions 6, 7, and 8 would be the most likely to produce a sample that is representative of the population.

Asking all neighbors would be the most likely to produce a representative sample.

10. Determine a better way to select a sample for this study.

Ask every 3rd person who walks by her house for 1 week. This would produce the most random, representative sample.